Department of Mechanical and Production Engineering

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Exercises

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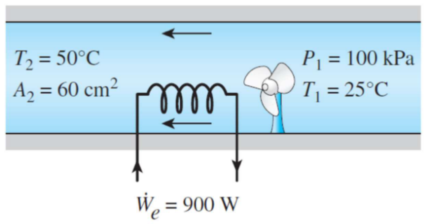
Contents

Problem 1–31

A hair dryer is basically a duct in which a few layers of electric resistors are placed. A small fan pulls the air in and forces it to flow over the resistors where it is heated. Air enters a 900 W hair dryer at 100 kPa and 25 °C, and leaves at 50 °C. The cross-sectional area of the hair dryer at the exit is 60 cm^2. Neglecting the power consumed by the fan and the heat losses through the walls of the hair dryer, determine



(a) the volume flow rate of air at the inlet and

(b) the velocity of the air at the exit.

Givens

$\dot{W}_e = 900\,\mathrm{W}$	Power consumption of the hair dryer
$P_1 = 100\,\mathrm{kPa}$	Pressure at the inlet
$T_1 = 25\,°\mathrm{C}$	Temperature at the inlet
$T_2 = 50\,°\mathrm{C}$	Temperature of the outlet
$A_2 = 60\,\mathrm{cm}^2$	Area of the outlet

Assumptions

- The air is an ideal gas
- The system is at steady state, $\frac{\mathrm{d}}{\mathrm{d}t} = 0$
- The fan does not consume power and no heat is lost through the wall
- c_p is assumed constant such that the enthalpy change is simply $c_p(T_2 - T_1)$
- The change in pressure and potential and kinetic energy of the air is negligible

Derivation

(a) Using the ideal gas law

$$p = \rho R T$$

we can solve for the density of the air at the inlet and outlet respectively, remembering we assume the pressure is the same at both places

$$\rho = \frac{p}{RT}$$
$$\rho_1 = \frac{100\,\mathrm{kPa}}{298.15\,\mathrm{K} \cdot 0.287\,\mathrm{kPa\,m^3/kg\,K}} = 1.169\,\mathrm{kg/m^3}$$
$$\rho_2 = \frac{100\,\mathrm{kPa}}{323.15\,\mathrm{K} \cdot 0.287\,\mathrm{kPa\,m^3/kg\,K}} = 1.078\,\mathrm{kg/m^3}.$$

As the entirety of the heat supplied by the heating element is assumed to go into the air stream the energy balance becomes

$$\dot{W}_e = \dot{m}c_p(T_2 - T_1) = \rho_1 Q_V c_p(T_2 - T_1).$$

We can now solve for the volume flow rate, Q_V as

$$Q_V = \frac{\dot{W}_e}{(T_2 - T_1)c_p \cdot \rho_1} = \frac{900\,\mathrm{W}}{25\,\mathrm{K} \cdot 1007\,\frac{\mathrm{J}}{\mathrm{kg\,K}} \cdot 1.169\,\mathrm{kg/m^3}} = 0.030\,58\,\mathrm{m^3/s}.$$

Result

The volume flow rate through the hair dryer is

$$Q_V = 0.031\,\mathrm{m^3/s}.$$

(b) We can use a similar formula as above, however, this time dividing by the outlet area as well to obtain the outlet flow speed:

$$V_2 = \frac{\dot{W}_e}{(T_2 - T_1)c_p \cdot \rho_2 \cdot A_2} = \frac{900\,\mathrm{W}}{25\,\mathrm{K} \cdot 1007\,\frac{\mathrm{J}}{\mathrm{kg\,K}} \cdot 1.078\,\mathrm{kg/m^3} \cdot 60\,\mathrm{cm^2}} = 5.526\,\frac{\mathrm{m}}{\mathrm{s}}.$$

Result

The outlet flow speed of the air is

$$V_2 = 5.526\,\frac{\mathrm{m}}{\mathrm{s}}.$$

Final result – Problem 1

(a) The volume flow rate through the hair dryer is

$$Q_V = 0.031\,\mathrm{m^3/s}.$$

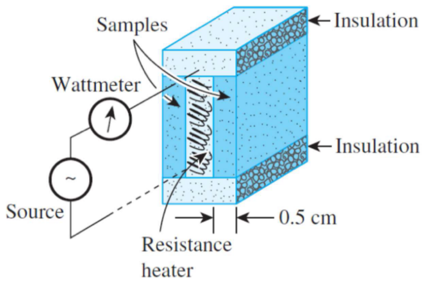
(b) The outlet flow speed of the air is

$$V_2 = 5.526\,\frac{\mathrm{m}}{\mathrm{s}}.$$

Problem 1–59

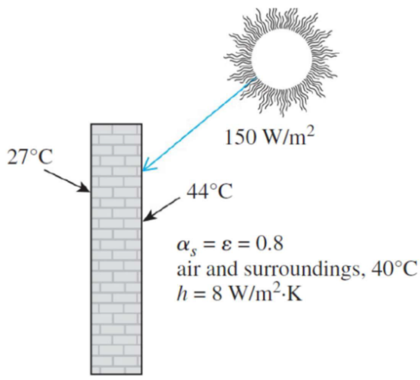
One way of measuring the thermal conductivity of a material is to sandwich an electric thermofoil heater between two identical rectangular samples of the material and to heavily insulate the four outer edges, as shown in the figure. Thermocouples attached to the inner and outer surfaces of the samples record the temperatures.

During an experiment, two 0.5 cm thick samples 10 cm × 10 cm in size are used. When steady operation is reached, the heater is observed to draw 25 W of electric power, and the temperature of each sample is observed to drop from 82 °C at the inner surface to 74 °C at the outer surface. Determine the thermal conductivity of the material at the average temperature.



Problem 1–100

The inner and outer surfaces of a 25 cm thick wall in summer are at 27 °C and 44 °C, respectively. The outer surface of the wall exchanges heat by radiation with surrounding surfaces at 40 °C, and convection heat transfer coefficient of 8 $\frac{\mathrm{W}}{\mathrm{m}^2\,\mathrm{K}}$. Solar radiation is incident on the surface at a rate of 150 $\frac{\mathrm{W}}{\mathrm{m}^2}$. If both the emissivity and the solar absorptivity of the outer surface are 0.8, determine the effective thermal conductivity of the wall.



Problem 1–106

A flat-plate solar collector is used to heat water by having water flow through tubes attached at the back of the thin solar absorber plate. The absorber plate has a surface area of $2\,\mathrm{m}^2$ with emissivity and absorptivity of 0.9. The surface temperature of the absorber is $35\,°\mathrm{C}$, and solar radiation is incident on the absorber at $500\,\frac{\mathrm{W}}{\mathrm{m}^2}$ with a surrounding temperature of $0\,°\mathrm{C}$. Convection heat transfer coefficient at the absorber surface is $5\,\frac{\mathrm{W}}{\mathrm{m}^2\,\mathrm{K}}$, while the ambient temperature is $25\,°\mathrm{C}$. Net heat rate absorbed by the solar collector heats the water from an inlet temperature (T_{in}) to an outlet temperature (T_{out}). If the water flow rate is $5\,\frac{\mathrm{g}}{\mathrm{s}}$ with specific heat of $4.2\,\frac{\mathrm{kJ}}{\mathrm{kg\,K}}$, determine the temperature rise of the water.